

STUDENT UTILITY PROGRAM

Java Programming Fundamentals – Lab Record

1. Objective

To develop a menu-driven Java Student Utility Program that demonstrates Java fundamentals such as input/output, conditional statements, switch statements, looping statements, parameterized methods, and basic problem-solving operations.

2. Task Description

Develop a menu-driven Student Utility Program. The program should allow the user to select different operations from a menu. The operations include student information and percentage calculation, even or odd checking, largest of three numbers, grade calculation, day of the week using switch, multiplication table, displaying numbers from 1 to N, sum of first N natural numbers, Fibonacci series, factorial, prime number checking, maximum of two numbers, and area of a circle. The program should also provide an exit option.

3. Step-wise Execution

Step 1: Import the Scanner class to accept input from the user.

Step 2: Create the StudentUtilityProgram class and define parameterized methods for factorial, prime checking, maximum, and circle area.

Step 3: Create the main() method and initialize the Scanner object.

Step 4: Display the Student Utility Program menu using System.out.println().

Step 5: Read the user's choice using nextInt().

Step 6: Use a switch statement to execute the selected operation.

Step 7: For student information, read name, roll number, three marks, calculate total and percentage, and display them.

Step 8: Use if-else for even/odd, largest of three numbers, and grade calculation.

Step 9: Use switch for converting a day number from 1 to 7 into the corresponding weekday.

Step 10: Use for loops for the multiplication table, numbers from 1 to N, sum, and Fibonacci series.

Step 11: Call the parameterized methods for factorial, prime checking, maximum, and circle area.

Step 12: Repeat the menu using a do-while loop until the user selects Exit (14).

Step 13: Close the Scanner object.

4. Algorithm

1. Start.
2. Create Scanner object.
3. Display the menu.
4. Read choice.
5. If choice is 1, read student details and calculate total and percentage.
6. If choice is 2, check whether the number is even or odd.
7. If choice is 3, find the largest among three numbers.
8. If choice is 4, determine grade from percentage using if-else.
9. If choice is 5, display the day using switch.
10. If choice is 6, print the multiplication table from 1 to 10.
11. If choice is 7, display numbers from 1 to N.
12. If choice is 8, calculate the sum of first N natural numbers.
13. If choice is 9, generate Fibonacci series.
14. If choice is 10, call factorial() and display the factorial.
15. If choice is 11, call prime() and display whether the number is prime.
16. If choice is 12, call maximum() and display the maximum.
17. If choice is 13, call areaCircle() and display the area.
18. If choice is 14, display Thank You and stop.
19. Otherwise, display Invalid Choice.
20. Repeat until choice equals 14.
21. Stop.

5. UML Diagram

StudentUtilityProgram	
Methods	Description
factorial(int n): long	Returns factorial of n
prime(int n): boolean	Checks whether n is prime
maximum(int a, int b): int	Returns larger number
areaCircle(double r): double	Returns area of circle
main(String[] args): void	Controls menu and program execution

UML relationship: StudentUtilityProgram contains the main() method and four static utility methods. The main() method calls the required utility methods based on the user's menu selection.


```

int n = sc.nextInt();

if (n % 2 == 0)
    System.out.println("Even Number");
else
    System.out.println("Odd Number");
break;

case 3:

    System.out.print("Enter Three Numbers: ");
    int a = sc.nextInt();
    int b = sc.nextInt();
    int c = sc.nextInt();

    if (a >= b && a >= c)
        System.out.println("Largest = " + a);
    else if (b >= c)
        System.out.println("Largest = " + b);
    else
        System.out.println("Largest = " + c);
    break;

case 4:

    System.out.print("Enter Percentage: ");
    double per = sc.nextDouble();

    if (per >= 90)
        System.out.println("Grade A");
    else if (per >= 75)
        System.out.println("Grade B");
    else if (per >= 60)
        System.out.println("Grade C");
    else if (per >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Fail");
    break;

case 5:

    System.out.print("Enter Day Number (1-7): ");
    int day = sc.nextInt();

    switch (day) {
        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:
            System.out.println("Wednesday");
            break;
        case 4:
            System.out.println("Thursday");
            break;
        case 5:
            System.out.println("Friday");
            break;
        case 6:
            System.out.println("Saturday");
            break;
        case 7:
            System.out.println("Sunday");
            break;
        default:
            System.out.println("Invalid Day");
    }
    break;

case 6:

    System.out.print("Enter Number: ");
    int num = sc.nextInt();

    for (int i = 1; i <= 10; i++) {
        System.out.println(num + " x " + i + " = " + (num * i));
    }
    break;

case 7:

    System.out.print("Enter N: ");
    int limit = sc.nextInt();

    for (int i = 1; i <= limit; i++) {
        System.out.print(i + " ");
    }
    System.out.println();
    break;

case 8:

```

```

        System.out.print("Enter N: ");
        int x = sc.nextInt();

        int sum = 0;

        for (int i = 1; i <= x; i++)
            sum += i;

        System.out.println("Sum = " + sum);
        break;

    case 9:

        System.out.print("Enter Number of Terms: ");
        int terms = sc.nextInt();

        int first = 0, second = 1;

        System.out.print(first + " " + second + " ");

        for (int i = 3; i <= terms; i++) {
            int next = first + second;
            System.out.print(next + " ");
            first = second;
            second = next;
        }
        System.out.println();
        break;

    case 10:

        System.out.print("Enter Number: ");
        int fact = sc.nextInt();

        System.out.println("Factorial = " + factorial(fact));
        break;

    case 11:

        System.out.print("Enter Number: ");
        int p = sc.nextInt();

        if (prime(p))
            System.out.println("Prime Number");
        else
            System.out.println("Not Prime");
        break;

    case 12:

        System.out.print("Enter Two Numbers: ");
        int n1 = sc.nextInt();
        int n2 = sc.nextInt();

        System.out.println("Maximum = " + maximum(n1, n2));
        break;

    case 13:

        System.out.print("Enter Radius: ");
        double r = sc.nextDouble();

        System.out.println("Area = " + areaCircle(r));
        break;

    case 14:
        System.out.println("Thank You");
        break;

    default:
        System.out.println("Invalid Choice");
    }
} while (choice != 14);

sc.close();
}
}

```

7. Output Screen

Sample Output 1 – Student Information

```
===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
...
13. Area of Circle
14. Exit
Enter your choice: 1
Enter Name: Ashiya
Enter Roll Number: 101
Enter Marks of Subject1: 85
Enter Marks of Subject2: 90
Enter Marks of Subject3: 80
Name : Ashiya
Roll No : 101
Total : 255
Percentage : 85.0
```

Sample Output 2 – Other Menu Operations

```
Enter your choice: 2
Enter Number: 24
Even Number
```

```
Enter your choice: 4
Enter Percentage: 88
Grade B
```

```
Enter your choice: 6
Enter Number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50
```

```
Enter your choice: 10
Enter Number: 5
Factorial = 120
```

```
Enter your choice: 11
Enter Number: 17
Prime Number
```

```
Enter your choice: 13
Enter Radius: 7
Area = 153.86
```

```
Enter your choice: 14
Thank You
```

Result

The Student Utility Program was successfully implemented in Java as a menu-driven application. It demonstrates input handling, selection statements, loops, and parameterized methods.